Econometrics Midterm

Multiple Choice · Econometrics 1–3 (Slides), Wooldridge Ch.1–6, Quiz 2 · 54 questions

1. An econometrician wants to learn the average hourly wage of all workers in Korea and collects data on 2,000 randomly selected workers. Which statement is correct?
   a. A sample is used only when the population is hypothetical
   b. The 2,000 workers are the population, because they are the units actually analyzed
   c. The 2,000 workers are the sample, and the estimates are statements about what the population of all Korean workers is like
   d. The estimates describe only the 2,000 workers and say nothing about other workers

2. Which statement best defines external validity?
   a. Whether the OLS estimator is unbiased within the sample used
   b. Whether the sample size is large enough to compute standard errors
   c. Whether the R-squared of the regression is high
   d. The validity of applying the conclusions of a study outside the context of that study

3. A researcher surveys only students at one private university in Seoul to estimate the average monthly smartphone spending of all Korean adults. What is the main concern?
   a. OLS cannot be computed because the population is too large
   b. The sample is unlikely to be representative of the population, so the conclusions may have weak external validity
   c. The estimates are automatically unbiased for all Korean adults because the students were chosen at random within the university
   d. The population becomes hypothetical, so no estimate is possible

4. Consider the model $\log(\text{price}) = \beta_0 + \beta_1 \text{subsidy} + \beta_2 \log(\text{sunlight}) + u$. How should β_1 be interpreted?
   a. The elasticity of price with respect to subsidy
   b. The slope from a simple regression of price on subsidy
   c. The change in log(price) from a one-unit increase in subsidy, holding log(sunlight) fixed (a partial slope)
   d. The total change in price from a one-unit increase in subsidy, including any change in sunlight it causes

5. In the model $\log(\text{price}) = \beta_0 + \beta_1 \text{subsidy} + \beta_2 \log(\text{sunlight}) + u$, which statement about subsidy is correct?
   a. Subsidy has a linear relationship with price, but not with log(price)
   b. Subsidy has a linear relationship with log(price), but not with price
   c. Subsidy has a linear relationship with both price and log(price)
   d. Subsidy has a linear relationship with neither price nor log(price)

6. In the model log(price) = $\beta_0 + \beta_1$ subsidy + β_2 log(sunlight) + u, which statement about sunlight is correct?

a. β_2 is the % change in price for a one-unit increase in sunlight

b. β_2 is the change in log(price) for a 1% increase in price

c. log(sunlight) has a linear relationship with log(price), so β_2 is approximately the % change in price for a 1% increase in sunlight, holding subsidy fixed

d. Sunlight has a linear relationship with price, so β_2 is the change in price for one more unit of sunlight

7. Suppose log(price) = $\beta_0 + \beta_1$ battery + β_2 battery^2 + u with $\beta_2 \neq 0$. Which statement is correct?

a. β_1 alone is the constant slope of battery for every value of battery

b. The model violates the assumption of linearity in parameters, so OLS cannot be used

c. Battery has a linear relationship with log(price) because the model is linear in parameters

d. Battery does not have a linear relationship with log(price); the effect of one more unit of battery depends on the current level of battery

8. Suppose log(price) = $\beta_0 + \beta_1$ battery + β_2 sunlight + β_3 battery× sunlight + u with $\beta_3 \neq 0$. Holding sunlight fixed, what is the effect on log(price) of one more unit of battery?

a. β_3· battery

b. β_1

c. $\beta_1 + \beta_3$

d. $\beta_1 + \beta_3$· sunlight

9. Which statement describes random sampling?

a. Every element has an equal probability of being assigned to the treatment or control group

b. The researcher chooses units that look typical of the population

c. Every element in the population has an equal probability of being selected into the sample

d. The sample contains more than half of the population

10. A researcher has a random sample of 500 workers and then flips a coin for each worker to decide who joins a job training program. The coin flip is an example of:

a. Random sampling

b. Perfect collinearity

c. Partialling out

d. Random assignment

11. Consider Model 2: wage = $\beta_0 + \beta_1$ educ + u, where ability is not included. People with better ability tend to get more education. Why can the OLS estimate of β_1 be biased even with a random sample?

a. There is no sample variation in educ

b. Ability goes into u and is correlated with educ, so the zero conditional mean condition E(u|educ) = 0 fails

c. Random sampling guarantees unbiasedness, so the estimate cannot be biased

d. The homoskedasticity assumption fails, which always causes bias

12. In Model 2, wage = $\beta_0 + \beta_1$ educ + u (ability omitted), suppose ability raises wage and is positively correlated with education. The OLS estimate of β_1 tends to:

a. Have a bias whose sign can never be determined
b. Understate the true return to education (downward bias)
c. Be unbiased, because ability is unobserved
d. Overstate the true return to education (upward bias)

13. The slides state: "OLS requires random sampling. It may also require random assignment." When estimating the effect of education on wages with Model 2 (wage = $\beta_0 + \beta_1$ educ + u), why would random assignment help?

a. Random assignment removes the need for any error term
b. Random assignment increases the R-squared of the regression
c. If educ were randomly assigned, it would be unrelated to ability and other factors in u, so the zero conditional mean condition would hold
d. Random assignment makes the sample representative of the population

14. A data set records the crime rate and the number of police officers for the same 150 cities in both 2020 and 2023. What type of data is this?

a. Pooled cross sections
b. Cross-sectional data
c. Panel (longitudinal) data
d. Time series data

15. A researcher draws a random sample of house prices in 1993 and a new, independent random sample of house prices in 1995 to evaluate a property tax change. What type of data is this?

a. Panel data
b. A single cross section
c. Time series data
d. Pooled cross sections

16. Which feature is typical of time series data but NOT of a cross section drawn by random sampling?

a. Observations are typically serially correlated, and their ordering conveys important information
b. Observations are more or less independent of each other
c. The ordering of observations does not matter
d. All observations are collected at the same point in time

17. In the simple regression y = $\beta_0 + \beta_1$ x + u, the OLS estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ are chosen to:

a. Minimize the sum of squared residuals
b. Make the fitted value equal to y for every observation
c. Minimize the sum of residuals
d. Minimize the sum of absolute residuals

18. In a simple regression of y on x, $R^2 = 0.25$. Which statement is correct?
    a. 25% of the observations lie exactly on the regression line
    b. 25% of the sample variation in y is explained by x
    c. x causes 25% of the change in y
    d. The estimated slope equals 0.25

19. The fitted regression is log(wage) = 0.584 + 0.083educ. How should 0.083 be interpreted?
    a. A 1% increase in education raises the wage by 8.3%
    b. One more year of education is associated with an approximately 8.3% higher wage
    c. One more year of education raises the wage by 0.083%
    d. One more year of education raises the wage by $0.083

20. Real earnings (in $1,000s) are regressed on train (= 1 if the person joined a randomly assigned job training program, 0 otherwise): earn = 4.55 + 1.79train. What does 1.79 measure?
    a. Nothing, because OLS cannot be used when x is binary
    b. Participants earn about 1.79% more than non-participants
    c. The average earnings of participants are $1,790
    d. The average treatment effect: participants earn about $1,790 more on average than non-participants

21. Under SLR.1–SLR.5, $\text{Var}(\hat{\beta}_1) = \sigma^2 / \text{SST}_x$. Which change makes $\hat{\beta}_1$ more precise (smaller variance)?
    a. More sample variation in x
    b. Fewer observations
    c. A larger error variance σ^2
    d. Replacing homoskedasticity with heteroskedasticity

22. In $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$, the OLS estimate $\hat{\beta}_1$ can also be obtained by:
    a. Regressing y on x_2 first, then adding x_1
    b. Regressing x_1 on x_2 (with a constant), then regressing y on the residuals from that regression
    c. Regressing x_2 on the residuals from a regression of y on x_1
    d. Averaging the slopes of the two simple regressions

23. Which regression violates MLR.3 (no perfect collinearity)?
    a. log(salary) on log(sales) and ceoten
    b. wage on educ and educ^2
    c. wage on educ and exper, where their correlation is 0.9
    d. wage on an intercept, male (= 1 if male) and female (= 1 if female)

24. The true model is $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$, but x_2 is omitted and y is regressed on x_1 only. In which case is the simple regression slope unbiased for β_1?
    a. When the sample size is very large
    b. When $\beta_2 = 0$ or when x_1 and x_2 are uncorrelated
    c. When the R-squared is high
    d. When x_2 is positively correlated with x_1 and $\beta_2 > 0$

25. Using 506 communities, the simple regression gives log(price) = 11.71 − 1.043 log(nox), while adding rooms gives log(price) = 9.23 − 0.718 log(nox) + 0.306 rooms. rooms raises price and is negatively correlated with log(nox). Which statement is correct?
    a. The simple regression is biased upward (toward zero), so −1.043 understates the true effect
    b. Adding rooms must always make the nox elasticity more negative
    c. The simple-regression elasticity −1.043 is biased downward (too negative), as predicted by $\beta_2 > 0$ and Corr(log(nox), rooms) < 0
    d. The two estimates differ only because of sampling error

26. High (but not perfect) correlation between x_1 and x_2 in a multiple regression mainly:
    a. Violates MLR.3, so OLS cannot be computed
    b. Increases the sampling variances of $\hat{\beta}_1$ and $\hat{\beta}_2$ but does not cause bias
    c. Causes $\hat{\beta}_1$ and $\hat{\beta}_2$ to be biased
    d. Inflates the variance of every coefficient, including an unrelated x_3

27. The Gauss–Markov theorem states that under MLR.1–MLR.5, the OLS estimators are:
    a. Exactly normally distributed in any sample
    b. The only unbiased estimators of the coefficients
    c. The best estimators even when the errors are heteroskedastic
    d. BLUE: they have the smallest variance among all linear unbiased estimators

28. In a regression, $\hat{\beta}_j = 0.80$ with $se(\hat{\beta}_j) = 0.10$. What is the t statistic for H_0: $\beta_j = 1$?
    a. 8.0
    b. −2.0
    c. 2.0
    d. −0.2

29. In a two-sided t test, the p-value is 0.07. Which statement is correct?
    a. H_0 is rejected at the 10% level but not at the 5% level
    b. H_0 is rejected at the 5% level
    c. H_0 is rejected at the 1% level
    d. The probability that H_0 is true is 7%

30. The 95% confidence interval for β_1 is [0.02, 0.10]. Which statement is correct?
    a. In repeated samples, intervals constructed this way contain the true β_1 95% of the time, and H_0: $\beta_1 = 0$ is rejected at the 5% level (two-sided)
    b. There is a 95% probability that $\hat{\beta}_1$ lies in [0.02, 0.10]
    c. β_1 equals 0.06 with 95% certainty
    d. H_0: $\beta_1 = 0$ cannot be rejected at the 5% level

31. To test H_0: $\beta_3 = 0$, $\beta_4 = 0$, $\beta_5 = 0$ jointly, which statement is correct?
    a. The F statistic can be negative when the restrictions help the fit
    b. Three separate t tests always give the same conclusion as the F test
    c. The restricted model always has a smaller SSR than the unrestricted model
    d. Use an F test comparing the SSR of the restricted and unrestricted models; F is large when the SSR rises a lot after imposing the restrictions

32. Which can cause the usual OLS t statistics to be invalid (not t-distributed under H_0)?
   a. A sample correlation of 0.95 between two explanatory variables in the model
   b. A low R-squared
   c. Heteroskedasticity, or omitting an important explanatory variable
   d. Having 500 observations instead of 5,000

33. For 32 chemical firms, rdintens = 0.472 + 0.321 log(sales) + 0.050 profmarg, where rdintens is R&D as a percentage of sales. If sales increase by 10%, rdintens changes by about:
   a. 3.21 percentage points
   b. 0.032 percentage points
   c. 32.1 percent
   d. 0.321 percentage points

34. Which statement about the consistency of OLS is correct?
   a. Consistency requires the errors to be normally distributed (MLR.6)
   b. Consistency requires homoskedasticity (MLR.5)
   c. A consistent estimator is unbiased in every finite sample
   d. Under MLR.1–MLR.4, the distribution of $\hat{\beta}_j$ collapses to the true β_j as $n \to \infty$, i.e. plim $\hat{\beta}_j = \beta_j$

35. pctstck = $\beta_0 + \beta_1$ funds + β_2 risktol + u satisfies MLR.1–MLR.4 (pctstck = % of pension in stocks, funds = number of mutual funds offered, risktol = risk tolerance). If funds and risktol are positively correlated and pctstck is regressed on funds only, the inconsistency in $\tilde{\beta}_1$ is:
   a. Zero, because the full model satisfies MLR.1–MLR.4
   b. Negative, because risktol is omitted
   c. It vanishes as the sample size grows
   d. Positive: plim $\tilde{\beta}_1 - \beta_1 = \beta_2\delta_1 > 0$, since $\beta_2 > 0$ and funds and risktol are positively correlated

36. cigs is the average number of cigarettes smoked per day. Can cigs have a normal distribution in the adult population, and what does this mean for OLS?
   a. Yes, by the central limit theorem every variable is normal
   b. No: cigs is nonnegative, piles up at zero (most adults do not smoke) and is discrete; still, with a large n, OLS t and F tests are approximately valid
   c. Yes, because cigs is an average
   d. No, so OLS estimates in a cigs equation are biased

37. Under MLR.1–MLR.5 but without normality (MLR.6), with a large sample:
   a. t tests become invalid no matter how large n is
   b. OLS becomes biased
   c. The usual t and F statistics are approximately valid, because $(\hat{\beta}_j - \beta_j)/\text{se}(\hat{\beta}_j)$ is approximately standard normal
   d. OLS standard errors cannot be computed

38. Under MLR.1–MLR.5, $\mathrm{se}(\hat{\beta}_j) \approx c_j / \sqrt{\{n\}}$. If the sample size is quadrupled, the standard error is approximately:

a. Quartered
b. Halved
c. Unchanged
d. Doubled

39. Crime model (n = 2,725): narr86 is regressed on pcnv, ptime86, qemp86 (restricted model). Its residuals are regressed on pcnv, avgsen, tottime, ptime86, qemp86, giving $R^2 = 0.0015$. To test H_0: avgsen and tottime have no effect:

a. $LM = nR^2 \approx 4.09$; compared with χ^2_2 (10% critical value 4.61), H_0 is not rejected at the 10% level
b. $LM = 0.0015$, so H_0 is rejected at the 1% level
c. $LM \approx 4.09$; compared with χ^2_5, H_0 is rejected
d. $LM = n(1 - R^2) \approx 2{,}721$, so H_0 is rejected

40. The dependent variable is changed from dollars to thousands of dollars (divided by 1,000). What happens?

a. Only the intercept changes
b. t statistics are divided by 1,000
c. R-squared falls
d. All coefficients and standard errors are divided by 1,000; t statistics and R-squared are unchanged

41. For 32 chemical firms, rdintens = 2.613 + 0.00030 sales − 0.0000000070 sales^2 (sales in $ millions). With salesbil = sales/1,000, the coefficients on salesbil and salesbil^2 become:

a. 0.00000030 and −0.0000000000070
b. 0.30 and −0.0000070
c. 0.30 and −0.0070, with the intercept and R-squared unchanged
d. 0.030 and −0.00070

42. For rdintens = 2.613 + 0.00030 sales − 0.0000000070 sales^2 (sales in $ millions), at what level of sales does the marginal effect of sales on rdintens become negative?

a. About 2,143
b. About 21,429, i.e. $21.4 billion
c. About 42,857
d. At sales = 0

43. A housing equation with dependent variable log(price) contains −0.545 rooms + 0.062 rooms^2 (other controls included). Moving from 6 to 7 rooms raises price by about:

a. 6.2%
b. 12.4%
c. −54.5%
d. 19.9%

44. In log(price) = 9.23 − 0.718 log(nox) + 0.306 rooms, what is the exact percentage change in price from one more room?

a. Exactly 30.6%

b. 0.306%

c. 3.06%

d. About 35.8%, = 100[exp(0.306) − 1]

45. In a regression where price and all explanatory variables are standardized (z-scores), the coefficient on nox is −0.340. Interpretation:

a. A 1 ppm increase in nox lowers price by 34%

b. A one standard deviation increase in nox lowers price by 0.34 standard deviation, holding the other variables fixed

c. A 1% increase in nox lowers price by 0.34%

d. nox explains 34% of the variation in price

46. log(wage) = 5.65 + 0.047educ + 0.00078educ· pareduc + 0.019exper + 0.010tenure (n = 722), where pareduc is total years of both parents' education. What is the estimated return to education when pareduc = 32?

a. 4.7%

b. 0.078%

c. About 7.2%

d. 2.5%

47. Which statement about adjusted R-squared is correct?

a. It always rises when a variable is added

b. It is always larger than R-squared

c. It can compare a model for y with a model for log(y)

d. It can fall when a variable is added; it rises only if the new variable's |t| exceeds 1

48. College GPA equation (n = 4,137): predicted colgpa = 2.70 at sat = 1200, hsperc = 30, hsize = 5, with se($\hat{y}^0$) = 0.020 and $\hat{\sigma}$ = 0.560. Which is correct?

a. The prediction interval for one student is narrower than the confidence interval for the mean

b. Both 95% intervals are 2.66 to 2.74

c. The prediction interval shrinks to zero width as n grows

d. The 95% CI for expected colgpa is about 2.66–2.74, but the 95% prediction interval for one student is about 1.60–3.80, because it also includes the error variance σ^2

49. The dependent variable is log(y). Using exp(predicted log(y)) as the prediction of y:

a. Systematically underestimates E(y|x); it should be scaled up by $\hat{\alpha}_0$, e.g. exp($\hat{\sigma}^2$/2) under normality or the average of exp($\hat{u}_i$)

b. Gives an unbiased prediction of y

c. Overestimates E(y|x)

d. Is valid only when R-squared is high

50. Palm oil production is estimated as: production = 243.63 − 3.34 tax − 0.61 fuel − 0.03 fuel^2. Production is the amount of palm oil produced, tax is the tax rate (%), and fuel is the fuel price ($). What is the predicted amount of palm oil produced when the tax rate is 5% and the fuel price is $7?

a. 222.45
b. 221.19
c. 226.93
d. 222.66

51. The palm oil equation above is estimated with an intercept and three regressors (tax, fuel, fuel^2). What are the degrees of freedom when the sample size is 85?

a. 84
b. 82
c. 81
d. 80

52. Worker productivity is estimated as: productivity = 3.36 + 0.06 log(training) + 2.53 log(experience) + 0.03 log(training) × log(experience). Productivity is shoes made per hour, training is training hours, experience is years of experience (log = natural log). What is the predicted number of shoes the worker makes in one hour with 2000 hours of training and 17 years of experience?

a. 6.79
b. 10.98
c. 11.63
d. 11.30

53. Vaccination is estimated as: log(vaccination) = 0.34 + 0.15 reminder + 0.23 log(ambassador). Reminder is the number of reminders sent and ambassador is the number of ambassadors who are information hubs (log = natural log). What is the predicted amount of log(vaccination) when 1 reminder is sent and there are five ambassadors?

a. 0.65
b. 0.86
c. 1.64
d. 0.49

54. Vaccination is estimated as: log(vaccination) = 0.34 + 0.15 reminder + 0.23 log(ambassador). Reminder is the number of reminders sent and ambassador is the number of ambassadors who are information hubs (log = natural log). What is the predicted amount of vaccination when 1 reminder is sent and there are five ambassadors?

a. 4.47
b. 0.86
c. 2.36
d. 5.16

Econometrics Midterm — Answers

1. c
A sample is the subset of the population chosen for analysis, and we study it to learn what the population is like. An estimate is a judgment about the population based on the sample.

2. d
External validity asks whether a study's conclusions can validly be applied outside the context of that study. It is directly tied to whether the sample represents the population.

3. b
Students at one university are unlikely to represent all Korean adults. We must ask whether the sample represents the population, and if it does not, the conclusions are hard to apply to that population (weak external validity).

4. c
In multiple regression β_1 is not a simple slope but a partial slope: the effect with the other regressor (log(sunlight)) held fixed. The dependent variable is log(price), so the units are also in terms of log(price).

5. b
log(price) is a linear function of subsidy, so that relationship is linear. But price = $\exp(\beta_0 + \beta_1$ subsidy + ...), so price and subsidy are not linearly related.

6. c
Log-log form: log(sunlight) and log(price) are linearly related, and β_2 is an elasticity (a 1% increase leads to about a β_2% change). Sunlight and price themselves are not linearly related.

7. d
Δ log(price)/Δ battery $\approx \beta_1 + 2\beta_2$ battery depends on the level of battery, so the relationship is not linear. The model is still linear in the parameters, so OLS can estimate it.

8. d
Because battery and sunlight interact, the effect of battery is $\beta_1 + \beta_3 \cdot$ sunlight and depends on the level of sunlight. So β_1 alone is not the slope of battery, and battery and log(price) are not linearly related.

9. c
Random sampling means every element of the population has the same probability of being drawn into the sample. Equal probability of group assignment is random assignment.

10. d
Drawing the sample is random sampling. Splitting the drawn people into treatment and control groups by a coin flip is random assignment.

11. b
If ability is left out of the model it goes into the error u. If ability and education are correlated, E(u|educ) = 0 (zero conditional mean) fails, so β_1 is biased even under random sampling.

12. d
Omitted variable bias = (effect of ability β_2) × (direction of the correlation between educ and abil). Both are positive, so the bias is positive, that is, the estimate is overstated.

13. c
With random assignment, education is not systematically related to factors in the error such as ability, so E(u|educ) = 0 holds and β_1 is unbiased. (Representativeness is the job of random sampling.)

14. c
Observing the same units (cities) repeatedly over time gives panel data. It lets us control for unobserved characteristics that do not change over time.

15. d
Combining two cross sections drawn independently at two points in time gives a pooled cross section. The same households are not followed, so it is not a panel. It is often used to compare before and after a policy change.

16. a
In a time series the observations are serially correlated and their order matters (trend, seasonality). In a randomly sampled cross section the observations are roughly independent and their order carries no information.

17. a
OLS (ordinary least squares) minimizes the sum of squared residuals $\Sigma\hat{u}_i^2$. As a result the residuals always sum to zero (an algebraic property).

18. b
R^2 = SSE/SST = 1 − SSR/SST is the proportion of the sample variation in y explained by x. It does not mean causality or the size of the slope.

19. b
Log-level (semi-log) form: a one-unit increase in x changes y by about $100\beta_1$ %. 0.083 × 100 = 8.3%.

20. d
If x is 0/1, the slope is the difference between the two group means. Under random assignment OLS is an unbiased estimator of the treatment effect. The non-participant mean is $4,550, so 1,790 is about a 39.3% increase.

21. a
The variance rises with the error variance σ^2 and falls with the sample variation in x, SST_x.

22. b
Partialling out: regress y on the residual of x_1 after removing the part explained by x_2 (the part uncorrelated with x_2), and the slope is β_1.

23. d
With an intercept, male + female = 1 is an exact linear relationship, so there is perfect collinearity. $educ^2$ is not a linear function of educ, and a high correlation (0.9) is multicollinearity, not a violation of MLR.3.

24. b
Omitted variable bias = $\beta_2 \times \delta_1$. If the omitted variable is irrelevant ($\beta_2 = 0$) or uncorrelated with x_1 ($\delta_1 = 0$), there is no bias.

25. c
Sign of the omitted variable bias = (effect of rooms β_2, +) × (correlation between log(nox) and rooms, −) = negative, so the bias is downward. Indeed −1.043 is more negative than −0.718. (Wooldridge problem, HPRICE2)

26. b
Multicollinearity only inflates the variances of the related variables (diagnosed with VIF) and unbiasedness is preserved. It is not a violation of MLR.3 in the strict sense.

27. d
Under MLR.1–5, OLS has the smallest variance among linear unbiased estimators (BLUE). With heteroskedasticity a better estimator exists. Normality requires MLR.6.

28. b
$t = (\hat{\beta}_j - a_j)/se(\hat{\beta}_j) = (0.80 - 1)/0.10 = -2.0$. Subtract the hypothesized value and divide by the standard error.

29. a
The p-value is the smallest significance level at which H_0 can be rejected. Reject only when $p < \alpha$: $0.07 < 0.10$ but $0.07 > 0.05$.

30. a
The confidence interval's endpoints are random (the parameter is not). Zero lies outside the 95% interval, so the two-sided 5% test rejects $\beta_1 = 0$.

31. d
$F = [(SSR_r - SSR_{ur})/q] \,/\, [SSR_{ur}/(n-k-1)]$. Imposing restrictions can only raise SSR, so $F \geq 0$, and the individual t tests and the joint test can give different conclusions.

32. c
Heteroskedasticity violates MLR.5 and omitting an important variable violates MLR.4 (zero conditional mean), so the t distribution does not hold. A high correlation (0.95) between regressors is not a CLM violation (only perfect collinearity is ruled out). (Wooldridge Problem 4.1)

33. b
Level-log form: $\Delta y \approx (\beta_1/100)(\%\Delta x) = (0.321/100) \times 10 \approx 0.032$. rdintens is measured in percent, so this is a percentage-point change. (Wooldridge problem, RDCHEM)

34. d
Consistency: as the sample grows the distribution of the estimator concentrates on the true value. MLR.1–MLR.4 suffice, and even the weaker MLR.4′ ($E(u) = 0$, $Cov(x_j, u) = 0$) is enough. Normality and homoskedasticity are not needed.

35. d
Inconsistency = $\beta_2 \times \delta_1$ with $\delta_1 = Cov(funds, risktol)/Var(funds)$. A higher risk tolerance means a larger stock share, so $\beta_2 > 0$, and the correlation is positive, giving a positive inconsistency. It does not vanish as the sample grows. (Wooldridge Problem 5.2)

36. b
The normal distribution is continuous and symmetric and can take negative values. cigs has a mass at zero, so it is not normal. But normality (MLR.6) is unrelated to unbiasedness, and in large samples inference is approximately valid thanks to asymptotic normality. (Wooldridge Problem 5.3)

37. c
Asymptotic normality (central limit theorem): even without normality, t and F tests and confidence intervals are approximately valid in large samples. Homoskedasticity (MLR.5) is still needed, though. With heteroskedasticity the usual standard errors are wrong even in large samples.

38. b
Standard errors shrink at the rate $1/\sqrt{n}$. Since $\sqrt{4} = 2$, they fall to about half.

39. a
LM = $n \cdot R^2_u = 2{,}725 \times 0.0015 \approx 4.09$, the degrees of freedom are the number of restrictions q = 2, and the p-value ≈ 0.13. Regress the restricted-model residuals on all of the regressors. (Wooldridge Example 5.3)

40. d
Changing the units of y changes the coefficients and standard errors by the same factor. So the t statistics, p-values and R^2 do not change (same statistical conclusions).

41. c
salesbil coefficient = 0.00030 × 1,000 = 0.30, and salesbil^2 = sales^2/1,000,000, so its coefficient = −0.0000000070 × 10^6 = −0.0070. This gives a form that is easy to report. (Wooldridge Problem 6.3)

42. b
Turning point x^* = $\hat{\beta}_1/(2|\hat{\beta}_2|)$ = 0.00030/(2 × 0.0000000070) ≈ 21,428.57 ($ millions). Beyond it the marginal effect becomes negative. (Wooldridge Problem 6.3)

43. d
%Δ price ≈ 100[−0.545 + 2(0.062)(6)] ≈ 19.9%. The turning point is 0.545/(2 × 0.062) ≈ 4.4 rooms, and only about 1% of the communities in the sample have fewer rooms than that. (Wooldridge Example 6.2, HPRICE2)

44. d
The 100β approximation for log-level is accurate only for small changes. The exact value is 100[exp(βΔ x) − 1] = 100(1.358 − 1) ≈ 35.8%.

45. b
A standardized (beta) coefficient shows how many standard deviations y changes when x changes by one standard deviation, so it lets us compare the effects of variables measured in different units. (Wooldridge Example 6.1)

46. c
Δ log(wage)/Δ educ = 0.047 + 0.00078 × pareduc = 0.047 + 0.025 ≈ 0.072. The higher the parents' education, the larger the return to education. (Wooldridge Problem 6.4, WAGE2)

47. d
Adjusted R^2 = 1 − [SSR/(n−k−1)]/[SST/(n−1)] penalizes through the degrees of freedom. R^2 never falls when variables are added, but adjusted R^2 can fall. It cannot be compared across different dependent variables (y vs log y).

48. d
Confidence interval for the mean: 2.70 ± 1.96(0.020). Prediction interval for one individual: se($\hat{e}^0$) = $\sqrt{\{0.020^2 + 0.560^2\}}$ ≈ 0.560, giving 2.70 ± 1.96(0.560). The error variance does not shrink as n grows. (Wooldridge Examples 6.5–6.6, GPA2)

49. a
E(y|x) = exp(σ^2/2)·exp(xβ), so using only exp(log ŷ) underpredicts. Multiply by $\hat{\alpha}_0 > 1$ to correct it. (Wooldridge Section 6.4, Example 6.7)

50. b
production = 243.63 − 3.34(5) − 0.61(7) − 0.03$(7)^2$ = 243.63 − 16.70 − 4.27 − 1.47 = 221.19.

51. c
df = n − k − 1 = 85 − 3 − 1 = 81.

52. c
ln(2000) = 7.6009 and ln(17) = 2.8332
productivity = 3.36 + 0.06(7.6009) + 2.53(2.8332) + 0.03(7.6009)(2.8332) = 3.36 + 0.4561 + 7.1680 + 0.6460 = 11.63
log base 10 would give 6.79, so log means ln.

53. b
log(vaccination) = 0.34 + 0.15(1) + 0.23 ln(5) = 0.34 + 0.15 + 0.23(1.6094) = 0.86.

54. c
vaccination = exp(0.8602) = $e^{0.8602}$ = 2.36.

Econometrics Midterm

Short Answer · Econometrics 1–3 (Slides), Wooldridge Ch.1–6, Quiz 2 · 37 questions

1. What is the difference between a population and a sample?

2. Why do econometricians usually study a sample instead of the population, and what is the goal of studying the sample?

3. Define external validity and state which property of the sample matters for it.

4. In $\log(\text{price}) = \beta_0 + \beta_1 \text{subsidy} + \beta_2 \log(\text{sunlight}) + u$, why is β_1 called a partial slope rather than simply "the slope"?

5. In $\log(\text{price}) = \beta_0 + \beta_1 \text{battery} + \beta_2 \text{sunlight} + \beta_3 \text{battery} \times \text{sunlight} + u$ with $\beta_3 \neq 0$, why does battery not have a linear relationship with log(price)?

6. What is the difference between random sampling and random assignment?

7. For Model 2, $\text{wage} = \beta_0 + \beta_1 \text{educ} + u$ (ability not included), explain why the zero conditional mean condition may be violated.

8. What is a control variable, and why is ability added in Model 1, wage = $\beta_0 + \beta_1$ educ + β_2 abil + u?

9. What does "ceteris paribus" mean, and why is it important in econometrics?

10. What is panel (longitudinal) data, and what is one advantage of it?

11. What does OLS minimize, and what is a residual?

12. What does R-squared measure in a regression?

13. What does the homoskedasticity assumption (SLR.5) state?

14. What does it mean that the OLS estimator β_1 is unbiased?

15. When does omitting a relevant-looking variable x_2 NOT cause bias in the estimated coefficient on x_1?

16. What is the main consequence of multicollinearity (high but not perfect correlation among regressors) for OLS?

17. State the Gauss–Markov theorem.

18. Define the p-value of a hypothesis test.

19. Does a statistically significant coefficient always matter in practice? Answer in 1–2 sentences.

20. Interpret a 95% confidence interval for β_j.

21. Which can make the usual OLS t statistics invalid (no t distribution under H_0): (i) heteroskedasticity, (ii) a sample correlation of 0.95 between two regressors in the model, (iii) omitting an important explanatory variable? Explain briefly.

22. What does it mean that an estimator is consistent, and which assumptions are enough for OLS to be consistent?

23. pctstck = $\beta_0 + \beta_1$ funds + β_2 risktol + u satisfies MLR.1–MLR.4 (risktol = risk tolerance). If funds and risktol are positively correlated and pctstck is regressed on funds only, what is the sign of the inconsistency in $\tilde{\beta_1}$, and why?

24. Can cigs (average cigarettes smoked per day) have a normal distribution in the adult population? What does this imply for OLS inference?

25. How is the LM (n-R-squared) statistic for testing q exclusion restrictions computed, and what distribution does it follow?

26. If the dependent variable is divided by 1,000 (dollars → thousands of dollars), what happens to the OLS coefficients, standard errors, t statistics and R-squared?

27. What is a beta (standardized) coefficient, and why is it useful?

28. For rdintens = 2.613 + 0.00030 sales − 0.0000000070 sales^2 (sales in $ millions), give the turning-point formula for a quadratic and find where the marginal effect of sales becomes negative.

29. In log(wage) = $\beta_0 + \beta_1$ educ + β_2 educ· pareduc + β_3 exper + β_4 tenure + u (pareduc = both parents' education), what is the return to one more year of education, and what does β_2 measure?

30. Why is adjusted R-squared used, and when does adding a variable increase it?

31. Why is a 95% prediction interval for one individual outcome y^0 much wider than the 95% confidence interval for $E(y|x = x^0)$?

32. When the dependent variable is log(y), why is exp(predicted log(y)) not a good prediction of y, and how is it fixed?

33. Palm oil production is estimated as: production = 243.63 − 3.34 tax − 0.61 fuel − 0.03 fuel^2. Production is the amount of palm oil produced, tax is the tax rate (%), and fuel is the fuel price ($). What is the predicted amount of palm oil produced when the tax rate is 5% and the fuel price is $7? Write the equation with the numbers substituted (two decimal places).

34. The palm oil equation above is estimated with an intercept and three regressors (tax, fuel, fuel^2). What are the degrees of freedom when the sample size is 85? Write the formula and the numbers.

35. Worker productivity is estimated as: productivity = 3.36 + 0.06 log(training) + 2.53 log(experience) + 0.03 log(training) × log(experience). Productivity is shoes made per hour, training is training hours, experience is years of experience (log = natural log). What is the predicted number of shoes the worker makes in one hour with 2000 hours of training and 17 years of experience? Write the equation with the numbers substituted (two decimal places).

36. Vaccination is estimated as: log(vaccination) = 0.34 + 0.15 reminder + 0.23 log(ambassador). Reminder is the number of reminders sent and ambassador is the number of ambassadors who are information hubs (log = natural log). What is the predicted amount of log(vaccination) when 1 reminder is sent and there are five ambassadors? Write the equation with the numbers substituted (two decimal places).

37. Vaccination is estimated as: log(vaccination) = 0.34 + 0.15 reminder + 0.23 log(ambassador). Reminder is the number of reminders sent and ambassador is the number of ambassadors who are information hubs (log = natural log). What is the predicted amount of vaccination when 1 reminder is sent and there are five ambassadors? Write the equation with the numbers substituted (two decimal places).

Econometrics Midterm — Model answers

1. Econ 1
A population is the entire set of individuals, objects or data points we want to study, while a sample is a subset of the population that is selected for analysis.
Key results: entire · subset · selected

2. Econ 1
Studying the whole population is often too costly or impossible, so we study a sample
the goal is to find out what the population is like from the sample.
Key results: costly · impossible · population is like

3. Econ 1
External validity is the validity of applying the conclusions of a study outside the context of that study
it depends on whether the sample is representative of the population.
Key results: outside · conclusions · representative

4. Econ 2
β_1 is the change in log(price) from a one-unit increase in subsidy while holding the other variables (log(sunlight)) fixed, so it isolates the effect of subsidy alone.
Key results: holding · other variables · one-unit

5. Econ 2
Because battery and sunlight interact, the effect of battery on log(price) is $\beta_1 + \beta_3 \cdot$sunlight, which depends on the level of sunlight
the slope is not constant.
Key results: interact · depends · not constant

6. Econ 3
In random sampling every element of the population has an equal probability of being selected
in random assignment every element has an equal probability of being assigned to a group.
Key results: equal probability · selected · assigned

7. Econ 3
Ability is omitted, so it goes into the error u
because people with better ability get more education, educ is correlated with u, so $E(u|educ) = 0$ fails and β_1 is estimated with bias.
Key results: ability · error · correlated

8. Econ 3
A control variable is included to hold another factor fixed so it no longer sits in the error
adding ability helps satisfy the zero conditional mean condition and removes the bias in β_1.
Key results: hold · zero conditional mean · bias

9. Ch.1
Ceteris paribus means "other relevant factors being equal"
most economic questions ask for the causal effect of one variable holding all other factors fixed.
Key results: other factors · equal · causal

10. Ch.1
Panel data follow the same cross-sectional units over time
they can account for time-invariant unobservables and model lagged responses.
Key results: same · over time · time-invariant

11. Ch.2
OLS chooses the estimates that minimize the sum of squared residuals
a residual is the actual value y_i minus the fitted value $\hat{y}_i$.
Key results: sum of squared residuals · actual · fitted

12. Ch.2
R^2 = SSE/SST is the fraction of the sample variation in y that is explained by the explanatory variable(s).
Key results: fraction · variation · explained

13. Ch.2
Homoskedasticity means the error u has the same (constant) variance for any value of x: $Var(u|x) = \sigma^2$.
Key results: variance · error · same

14. Ch.2
Unbiasedness means $E(\hat{\beta}_1) = \beta_1$: if sampling and estimation were repeated many times, the estimates would on average equal the true population value, though one estimate may differ.
Key results: on average · repeated · true

15. Ch.3
There is no omitted variable bias if the omitted variable is irrelevant (its coefficient is zero) or uncorrelated with x_1.
Key results: irrelevant · uncorrelated

16. Ch.3
Multicollinearity increases the sampling variance of the affected slope estimators, but OLS remains unbiased
it can be detected with variance inflation factors.
Key results: variance · increase · unbiased

17. Ch.3
Under MLR.1–MLR.5, OLS is BLUE (best linear unbiased estimator): it has the smallest variance among all linear unbiased estimators.
Key results: BLUE · smallest variance · MLR.1

18. Ch.4
The p-value is the smallest significance level at which the null hypothesis would be rejected
we reject H_0 when the p-value is below the chosen level.
Key results: smallest · significance level · reject

19. Ch.4
Not necessarily: statistical significance only shows the effect differs from zero, while economic (practical) significance depends on the magnitude of the coefficient.
Key results: not necessarily · magnitude · economic

20. Ch.4
In repeated samples, intervals built this way contain the true population coefficient in 95% of cases
the bounds are random, not the parameter.
Key results: repeated · contain · true

21. Ch.4
Heteroskedasticity (i) and an omitted important variable (iii) violate the CLM assumptions, so the t statistics are invalid
a high correlation of 0.95 (ii) is not a CLM violation (only perfect collinearity is ruled out).
Key results: heteroskedasticity · omitted · correlation

22. Ch.5
An estimator is consistent if, as the sample size grows, its distribution collapses to the true parameter (plim $\hat{\beta}_j = \beta_j$)
OLS is consistent under MLR.1–MLR.4 (even zero correlation between each x_j and u is enough).
Key results: sample size · collapse · true · MLR.4

23. Ch.5
The inconsistency is positive: plim $\tilde{\beta}_1 - \beta_1 = \beta_2\delta_1$, with $\beta_2 > 0$ (higher risktol raises pctstck) and $\delta_1 > 0$ because funds and risktol are positively correlated
it does not disappear as n grows.
Key results: positive · risktol · correlated · not disappear

24. Ch.5
No: most adults smoke zero cigarettes and cigs is nonnegative and discrete, so it is not normal
with a large sample, asymptotic normality still makes the t and F tests approximately valid.
Key results: zero · nonnegative · not normal · large sample

25. Ch.5
Estimate the restricted model, regress its residuals on all the explanatory variables, and compute LM = nR^2 from that auxiliary regression
under H_0 it is approximately chi-square with q degrees of freedom.
Key results: restricted · residuals · nR · chi-square

26. Ch.6
All coefficients and standard errors are divided by 1,000, while the t statistics and R-squared stay unchanged
only the units change, not the statistical conclusions.
Key results: divided · unchanged · t statistic · R-squared

27. Ch.6
A beta coefficient comes from regressing the standardized y on standardized x's
it is the change in y, in standard deviations, from a one standard deviation increase in x_j, so effects measured in different units can be compared.
Key results: standardized · standard deviation · units · compare

28. Ch.6
Turning point $\hat{x}^* = \hat{\beta}_1/(2|\hat{\beta}_2|) = 0.00030/(2 \times 0.0000000070) \approx 21{,}428.57$, i.e. sales of about $21.4 billion beyond that the estimated marginal effect of sales on rdintens is negative.

29. Ch.6
The return is $\beta_1 + \beta_2 \cdot$pareduc, so it depends on parents' education
the interaction coefficient β_2 is how much the return changes for one more year of pareduc (estimate 0.00078: at pareduc = 32 the return is about 7.2%).
Key results: depends · parents · interaction

30. Ch.6
R-squared never falls when a variable is added, so adjusted R-squared imposes a penalty for extra regressors using degrees of freedom
adding a variable raises it only if that variable's t statistic exceeds one in absolute value.
Key results: never falls · penalty · degrees of freedom · t statistic

31. Ch.6
The confidence interval reflects only the sampling variance of $\hat{y}^0$, while the prediction interval also includes the variance of the unobserved error for an individual outcome (σ^2), which does not shrink as n grows (e.g. colgpa: CI 2.66–2.74 vs PI 1.60–3.80).
Key results: error · variance · individual

32. Ch.6
$\exp(\log \hat{y})$ systematically underestimates $E(y|x)$
we multiply it by $\hat{\alpha}_0 = \exp(\sigma^2/2)$ under normal errors, or by the average of $\exp(\hat{u}_i)$ from the residuals.
Key results: underestimate · multiply · residuals

33. Ch.6
production $= 243.63 - 3.34(5) - 0.61(7) - 0.03(7)^2$
$= 243.63 - 16.70 - 4.27 - 1.47 = 221.19$
Key results: 221.19 · 16.7 · 4.27 · 1.47

34. Ch.3
df $= n - k - 1 = 85 - 3 - 1 = 81$
Key results: 81 · n − k − 1 · 85 − 3 − 1

35. Ch.6
ln(2000) $= 7.6009$ and $\ln(17) = 2.8332$
productivity $= 3.36 + 0.06(7.6009) + 2.53(2.8332) + 0.03(7.6009)(2.8332)$
$= 3.36 + 0.4561 + 7.1680 + 0.6460 = 11.63$
log base 10 would give 6.79, so log means ln
Key results: 11.63 · 7.60 · 2.83 · 7.17 · 0.65

36. Ch.6
log(vaccination) $= 0.34 + 0.15(1) + 0.23 \ln(5) = 0.34 + 0.15 + 0.23(1.6094) = 0.86$
Key results: 0.86 · 0.37 · 1.61

37. Ch.6
vaccination $= \exp(0.8602) = e^{0.8602} = 2.36$
Key results: 2.36 · 0.86

Econometrics Midterm

R functions and output interpretation · 5 code exhibits · W2, W5, W6 · 23 questions

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Exhibit A · Lab2.R — descriptive statistics (W2)

Used in questions 1–4

```
1  practice <- read.csv("Lab2_Practice.csv", header=TRUE, na.strings=".")
2  competence <- practice$COMPETENCE
3  breaks <- practice$BREAKS
4  grade <- practice$GRADE
5  hist(competence, breaks=15)
6  summary(breaks)
7  sum((competence-mean(competence))^2)/(length(competence)-1)
8  var(competence)
9  quantile(breaks)
10 IQR(breaks)
11 boxplot(grade)
12 abline(h=median(grade), col="red")
13 par(mfrow = c(1, 2))
14 boxplot(grade)
15 vioplot(grade)
```

---

1. [Exhibit A] Line 9. What does quantile() do?

2. [Exhibit A] Line 13. What does par(mfrow = c(1, 2)) do?

3. [Exhibit A] Lines 7–8. Both lines print 4.772. Why do they agree, and why does the formula divide by n − 1 and not by n?

4. [Exhibit A] Lines 9–10 (with line 6). The output for breaks is below. What do the mean and the median say about the shape of the distribution? Is the maximum an outlier by the 1.5 × IQR rule? Write the fence.

summary(breaks) and IQR(breaks) · n = 27

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	IQR
0	1	2	3.185	4.5	12	3.5

Exhibit B · Lab4.R — functions and strings (W2)

Used in questions 5–6

```
1  library(readxl)
2  library(stringr)
3  to12 <- function(str){
4    hour=as.integer(str)
5    if (hour>12) {
6      return(paste(hour-12, "PM"))
7    } else {
8      return(paste(hour, "AM"))
9    }
10 }
11 hasMultipleLoc <- function(str){
12   str_detect(str, fixed(","))
13 }
14 data=read_excel("Café.xlsx")
15 data <- as.data.frame(data)
16 colnames(data)=c("type","location","openingHour","closingHour")
17 (openingHour <- sapply(data$openingHour, to12))
18 data$openingHour=openingHour
19 (locationText <- ifelse(sapply(data$location, hasMultipleLoc), "Yes", "No"))
20 data$hasMultipleLoc=locationText
```

5. [Exhibit B] Line 12. What does str_detect() do, and what is fixed() for?

6. [Exhibit B] Line 17. What does sapply() do?

Exhibit C · regression.R (condensed) — OLS, inference and prediction (W5)

Used in questions 7–12

```
1  social <- read.csv("social.csv", header=T)
2  perform=social$cryp_scoreFINAL
3  eye=social$RMET_mean
4  skill=social$Smean_mean
5  ravens=social$ravens_mean
6  plot(perform ~ eye)
7  eye_regr=lm(perform ~ eye)
8  abline(eye_regr, col="blue")
9  abline(h=mean(perform), col="red")
10 abline(v=mean(eye), col="green")
11 skill_regr=lm(perform ~ skill)
12 skill_summary=summary(skill_regr)
13 confint(skill_regr, level=0.99)
14 (skill_coef=cor(perform,skill)^2)
15 round(skill_summary$r.squared-skill_coef,16)
16 mult_lm=lm(perform ~ skill + ravens)
17 (mult_summary=summary(mult_lm))
18 predict(mult_lm, data.frame(skill=c(0.5,1.5), ravens=c(1.5,0.5)), interval="confidence",
   level=0.95)
19 points <- seq(min(eye)-3, max(eye)+3, length.out=10000)
20 pred <- predict(eye_regr, data.frame(eye=points), interval="confidence", level=0.95)
21 polygon(c(rev(points), points), c(rev(pred[ ,3]), pred[ ,2]), col=rgb(0,0,0,0.1),
   border=NA)
```

7. [Exhibit C] Line 13. What does confint() do, and what does level = 0.99 change?

8. [Exhibit C] Line 18. What does predict() do, and what does interval = "confidence" add?

9. [Exhibit C] Lines 11–13. Read the output of `summary(skill_regr)` and `confint()` below (all variables are standardized). Interpret the slope, say whether it is significant at 5% and at 1% and how the two intervals show this, and say what share of the variation in perform skill explains.

summary(skill_regr) and confint() · n = 85

Term	Estimate	Std. Error	t value	p-value	95% CI	99% CI
(Intercept)	0.0000	0.1066	0.000	1.0000	[−0.212, 0.212]	[−0.281, 0.281]
skill	0.2141	0.1072	1.997	0.0491	[0.0009, 0.4274]	[−0.0685, 0.4968]
Multiple R-squared 0.04585 · Adjusted R-squared 0.03436 · F 3.989 on 1 and 83 DF, p = 0.0491						

10. [Exhibit C] Lines 16–17. Read the output of `summary(mult_lm)` below. Interpret the skill coefficient. Then explain how neither slope is significant at 5% on its own while the regression as a whole is.

summary(mult_lm) · n = 85

Term	Estimate	Std. Error	t value	p-value
(Intercept)	0.0000	0.1056	0.000	1.000
skill	0.1712	0.1096	1.562	0.122
ravens	0.1746	0.1096	1.593	0.115
Multiple R-squared 0.07449 · Adjusted R-squared 0.05192 · F 3.30 on 2 and 82 DF, p = 0.04184				

11. [Exhibit C] Lines 11 and 16. The skill slope is 0.2141 in the simple regression (line 11) and 0.1712 once ravens is added (line 16). The slope from regressing ravens on skill is 0.2458. Show how these numbers fit together and explain the drop.

12. [Exhibit C] Line 18. The output below also shows `interval = "prediction"` for the same two new teams. Interpret the first confidence interval, and explain why the prediction interval is so much wider.

predict() · 95%

New team	fit	conf. lwr	conf. upr	pred. lwr	pred. upr
skill 0.5, ravens 1.5	0.3475	−0.0339	0.7289	−1.6267	2.3217
skill 1.5, ravens 0.5	0.3441	−0.0373	0.7255	−1.6301	2.3183

Exhibit D · simple regression demonstration.R (condensed) — simulation of tests and intervals (W5)

Used in questions 13–17

```
1  set.seed(4201)
2  slope=0.2; intercept=3; sampleSize=10; sampleCnt=10000
3  estimator=c(); t=c(); pValue=c()
4  for(rowIdx in 1:sampleCnt) {
5    fertilizer=rnorm(sampleSize,10,2)
6    yield=intercept+slope*fertilizer+rnorm(sampleSize,0,1)
7    coef=summary(lm(yield~fertilizer))$coefficients
8    estimator[rowIdx]=coef[2,1]
9    t[rowIdx]=coef[2,3]
10   pValue[rowIdx]=coef[2,4]
11 }
12 hist(estimator)
13 alpha=0.05
14 rejection=ifelse(pValue<=alpha, "Reject", "Fail to Reject")
15 (result=table(rejection))
16 barplot(result, main="Null Hypothesis Rejection")
17 sampleSize=12; slope=2; sampleCnt=20
18 results=data.frame(index=1:sampleCnt)
19 for(rowIdx in 1:sampleCnt) {
20   fertilizer=rnorm(sampleSize,10,2)
21   yield=intercept+slope*fertilizer+rnorm(sampleSize,0,1)
22   fer_regr=lm(yield ~ fertilizer)
23   results[rowIdx,"slope"]=fer_regr$coefficients[2]
24   interval=confint(fer_regr)
25   lowerBound=interval["fertilizer","2.5 %"]
26   upperBound=interval["fertilizer","97.5 %"]
27   results[rowIdx,"inside"]=lowerBound<=slope && slope<=upperBound
28 }
```

13. [Exhibit D] Line 1. What does set.seed() do?

14. [Exhibit D] Line 14. What does ifelse() do?

15. [Exhibit D] Lines 4–12. Over the 10,000 samples, mean(estimator) = 0.2027 and sd(estimator) = 0.190, while the true slope is 0.2. What property of OLS does this show, and what does the histogram on line 12 picture?

16. [Exhibit D] Line 15. The table prints Fail to Reject 8092 and Reject 1908, and the true slope is 0.2, not 0. What does the 19.1% rejection rate measure, what is the other 80.9%, and what would the rate be if the true slope were 0?

table(rejection) · 10,000 samples

Fail to Reject	Reject
8092	1908

17. [Exhibit D] Lines 17–28. In this run all 20 intervals contain the true slope 2 (inside = TRUE 20 times). How many would you expect, and what does the 95% level actually promise?

Exhibit E · W6 — dummy variables and interactions (lecture examples, Wooldridge Ch.7)

Used in questions 18–23

```
1 m1 <- lm(wage ~ female + educ, data = df)
2 m2 <- lm(wage ~ educ*female, data = df)
3 m3 <- lm(wage ~ educ + as.factor(region), data = df)
4 library(car)
5 linearHypothesis(m2, c("female = 0", "educ:female = 0"))
6 restricted <- lm(wage ~ educ + exper, data = df)
7 unrestricted <- lm(wage ~ (educ + exper)*female, data = df)
8 anova(restricted, unrestricted)
9 lpm <- lm(inlf ~ educ + age, data = df)
```

---

18. [Exhibit E] Line 3. What does as.factor() do, and what does lm() then create from it?

19. [Exhibit E] Line 8. What does anova() do when it is given two models?

20. [Exhibit E] Line 1, estimated with df = wage1 (Wooldridge, n = 526), gives predicted wage = 0.623 − 2.273 female + 0.506 educ, with a standard error of 0.279 on female. Interpret the female coefficient. Which group is the base, and why can't a male dummy be added as well?

21. [Exhibit E] Lines 2 and 5. The output of m2 (wage1) is below. What is the estimated return to education for men and for women, and is the gap significant? Line 5 tests female = 0 and educ:female = 0 together and gives F = 33.51, p < 0.001. Why can it reject when neither t-test does?

summary(m2) · wage1, n = 526

Term	Estimate	Std. Error	t value	p-value
(Intercept)	0.2005	0.8436	0.238	0.812
educ	0.5395	0.0642	8.400	<0.001
female	−1.1985	1.3250	−0.905	0.366
educ:female	−0.0860	0.1036	−0.830	0.407

22. [Exhibit E] Line 8. The anova() table (wage1) is below. Compute F from the RSS values and state the conclusion. What does the test assume about the error variance?

anova(restricted, unrestricted) · wage1

Model	Res.Df	RSS	Df	Sum of Sq	F	p-value
restricted	523	5548.2				
unrestricted	520	4737.8	3	810.38	29.648	<2.2e-16

23. [Exhibit E] Line 9 is a linear probability model (inlf = 1 if a woman is in the labor force). The textbook version (Example 7.12, mroz, n = 753) is below. Interpret the educ and kidslt6 coefficients, and name two weaknesses of the LPM that this fit shows.

```
lpm <- lm(inlf ~ nwifeinc + educ + exper + I(exper^2) + age + kidslt6 + kidsge6,
data = mroz)
```

Selected coefficients · fitted values from −0.345 to 1.127

Term	Estimate	Std. Error	t value
educ	0.0380	0.0074	5.15
kidslt6	−0.2618	0.0335	−7.81
age	−0.0161	0.0025	−6.48
nwifeinc	−0.0034	0.0014	−2.35

Econometrics Midterm — Model answers

1. W2 · quantile()
quantile(): returns sample quantiles, the values below which given shares of the data fall
By default it gives five numbers, the 0%, 25%, 50%, 75% and 100% points: the **minimum**, the first **quartile**, the median, the third quartile and the **maximum**
Other points can be asked for with probs, for example quantile(breaks, probs = 0.9)
Key results: quartile · 0% · minimum

2. W2 · par(mfrow =)
par(): sets **graphics parameters** for the plots that follow
mfrow = c(1, 2) splits the plotting window into **1 row and 2 columns**, filled row by row
So the next two plots (lines 14–15) appear **side by side**, until the layout is reset with par(mfrow = c(1, 1))
Key results: graphics parameter · 1 row and 2 columns · side by side

3. W2 · Ch.2 sample variance
Line 7: computes the sample variance by hand, $s^2 = \Sigma(x_i - \bar{x})^2 \div (n - 1) = 4.772$ for competence (n = 27)
var(): uses exactly the same formula, so both lines print **4.772**
Why n − 1: $\bar{x}$ is estimated from the same data and uses up one **degree of freedom**, and dividing by n − 1 makes s^2 an **unbiased** estimator of σ^2
Same idea in Ch.3: $\sigma^2 = SSR \div (n - k - 1)$
Key results: 4.772 · var() · n - 1 · unbiased

4. W2 · Ch.2 output: summary() and IQR()
Mean vs median: mean 3.185 > median 2, so a few large values pull the mean up and the distribution is **right-skewed**
IQR = Q3 − Q1 = 4.5 − 1 = 3.5
Upper fence = Q3 + 1.5 × IQR = 4.5 + 1.5 × 3.5 = **9.75**
Max 12 > 9.75, so 12 is an **outlier** and boxplot() draws it as a separate point
Robustness: the median and the IQR are not moved by it, the mean and the variance are
Key results: right-skewed · 9.75 · outlier

5. W2 · str_detect()
str_detect(): checks whether each string **contains** a pattern and returns **TRUE** or FALSE for each one
fixed(): makes the pattern a **literal** string, not a regular expression
So hasMultipleLoc("Seoul, Suwon") is TRUE and hasMultipleLoc("Seoul") is FALSE
Key results: TRUE · contains · literal

6. W2 · sapply()
sapply(): **applies** a function to **each element** of a vector or list
It then simplifies the results into a **vector** (or a matrix) when it can
Here it runs to12() on every opening hour, which to12() cannot do by itself because its if() takes one value at a time
Key results: each element · applies · vector

7. W5 · confint()
confint(): returns a **confidence interval** for each **coefficient** of a fitted model, a lower and an upper bound per term
By default the level is 95%
level = 0.99 gives 99% intervals, which are **wider**
Key results: confidence interval · coefficient · wider

8. W5 · predict()
predict(): plugs **new data**, a data frame with the model's variable names, into the fitted equation and returns the **predicted** values, here about 0.3475 and 0.3441
interval = "confidence" adds the **lower and upper** bounds, so the result has the columns fit, lwr and upr
Key results: predicted · new data · lower

9. W5 · Ch.4 output: t-test, CI and R^2
Slope: one standard deviation more social skill goes with **0.214** SD higher team performance
t = 0.2141 ÷ 0.1072 = **1.997** with df = 85 − 2 = 83
p = **0.049** < 0.05, so H_0: $\beta_1 = 0$ is rejected at 5% but not at 1% (p > 0.01)
Intervals: the 95% CI [0.0009, 0.4274] excludes 0 and the **99%** CI [−0.0685, 0.4968] includes 0, because a two-sided test at level α rejects exactly when the 100(1 − α)% CI excludes the null value
R^2 = 0.04585: skill explains about **4.6%** of the variation, and in simple regression R^2 = cor(perform, skill)^2, which is why lines 14–15 print a difference of 0
Low R^2: does not make the slope useless, it can still be a precise ceteris paribus estimate (Ch.3)
Key results: 0.214 · 1.997 · 0.0491 · 99% · 4.6%

10. W5 · Ch.4 output: separate t-tests vs the F-test
Skill: holding ravens fixed (**ceteris paribus**), one SD more social skill goes with 0.171 SD higher performance
t-tests: p = **0.122** (skill) and 0.115 (ravens), so neither is significant at 5% on its own
F-test: H_0: $\beta_1 = \beta_2 = 0$ against H_1: at least one ≠ 0 gives F(2, 82) = 3.30 with p = **0.042** < 0.05, so the two are **jointly** significant
Why both can hold: each t-test asks whether one variable adds anything once the other is already in, and skill and ravens share explanatory power (correlation 0.246), so each looks weak alone while together they clearly explain performance
Adjusted R^2 = 0.0519 > 0.0344 (skill alone): adding ravens is worth the lost degree of freedom, as expected because its |t| = 1.59 > 1 (Ch.6)
Key results: holding ravens fixed · 0.122 · 0.0418 · jointly

11. W5 · Ch.3 omitted variable bias
Formula: simple slope = $\hat{\beta}_1 + \hat{\beta}_2 \times \tilde{\delta}_1$ (Ch.3)
Numbers: 0.1712 + 0.1746 × 0.2458 = 0.1712 + 0.0429 = **0.2141**
Why: ravens raises performance ($\hat{\beta}_2$ = **0.1746** > 0) and is positively correlated with skill ($\tilde{\delta}_1$ = **0.2458** > 0), so without ravens the skill slope picks up part of the ravens effect
Result: the simple slope is biased **upward** by 0.043, the **omitted variable** bias
Note: $\tilde{\delta}_1$ equals the correlation here because both variables are standardized
Key results: 0.2141 · 0.1746 · 0.2458 · omitted variable · upward

12. W5 · Ch.6 output: confidence vs prediction interval
Fit: $\hat{y}$ = 0.3475 for skill = 0.5 and ravens = 1.5
Confidence interval: [−0.034, 0.729] is a 95% range for the **average** performance E(y | x) of all teams with these scores, and it reflects only the uncertainty in the estimated coefficients
Prediction interval: [−1.627, 2.322] is a 95% range for **one** new team, and it adds the variance of that team's own **error** u, se($\hat{y}$)^2 + $\hat{\sigma}^2$ with $\hat{\sigma}$ = 0.974
Why wider: $\hat{\sigma}^2$ dominates and does not shrink as n grows, so the prediction interval is always **wider** (Ch.6)
Key results: 0.3475 · average · individual · error · wider

13. W5 · set.seed()
set.seed(): sets the starting point (seed) of the **random number generator**
Every run with the same seed draws the **same random numbers**, so the simulation is **reproducible**
Without it, rnorm() would give different samples on each run
Key results: reproducible · random number generator · same random numbers

14. W5 · ifelse()
ifelse(): checks a **condition** for **each element** of a vector
Where it is TRUE it returns the **second** argument, and where it is FALSE the third
Here every p-value becomes "Reject" (p-value at most 0.05) or "Fail to Reject"
Key results: condition · each element · second

15. W5 · Ch.3 unbiasedness in the simulation
Mean: 0.2027 ≈ 0.2, so across repeated samples OLS is right on average, $E(\hat{\beta}_1) = \beta_1$: **unbiasedness** (MLR.1–MLR.4)
hist(estimator): pictures the **sampling distribution** of $\hat{\beta}_1$, centred at 0.2 and roughly normal because the errors are normal (MLR.6)
Spread: sd 0.190 is the sampling standard deviation, and it shrinks with a larger n or more variation in fertilizer, since $Var(\hat{\beta}_1) = \sigma^2 \div SST_x$
Key results: unbiased · 0.2027 · sampling distribution

16. W5 · Ch.4 output: rejection rate = power
Power: H_0 ($\beta_1 = 0$) is false here, so every rejection is correct and 1908 ÷ 10,000 = **19.1%** estimates the **power** of the 5% t-test
Type II: the other 80.9% fail to reject a false H_0, a **Type II** error
Why low: the effect is small relative to the noise and n is only 10, so a larger slope or a larger n raises power
True slope 0: H_0 is then true and the rejection rate is about α = **5%**, the Type I error rate (a rerun with slope = 0 gave 5.2%)
Key results: power · 19.1% · type II · 5%

17. W5 · Ch.4 output: what "95% confident" means
Expected: 0.95 × 20 = **19** intervals contain the true slope, and 20 of 20 is well within normal sampling variation
Meaning: 95% is a property of the method, in **repeated samples** 95% of the intervals built this way cover the **true slope**
Single interval: once computed it either contains 2 or it does not, so "a 95% probability that β_1 lies in this interval" is the wrong reading
Key point: the bounds are **random**, β_1 is fixed (Ch.4)
Key results: 19 · repeated samples · true slope · random

18. W6 · as.factor()
as.factor(): converts a variable into a **factor**, a **categorical** variable with levels, instead of a number
lm(): then creates one **dummy** for each level except the **base** (reference) level
Each dummy coefficient is the difference from the base level
Key results: categorical · dummy · base

19. W6 · anova()
anova(): **compares** two **nested** models with an **F-test**
It uses the residual sum of squares (**RSS**) of each model and the number of extra parameters, and reports F and its p-value
Here the models differ by female and its interactions, so it is the **Chow** test
Key results: nested · F-test · RSS · Chow

20. W6 · Ch.7 output: dummy coefficient
female: among people with the **same education**, women earn about **$2.27** per hour less than men on average (t = −2.273 ÷ 0.279 = −8.15, p < 0.001)
Base group: men (female = 0), the intercept 0.623 belongs to the men's line and −2.273 shifts it down for women
Male dummy: female + male = 1 for everyone, an exact linear relation with the intercept, the **dummy variable trap** (perfect collinearity, MLR.3)
Caution: this is not proof of discrimination, since female may be correlated with omitted productivity factors such as experience (Ch.7)
Key results: 2.27 · same education · base · dummy variable trap

21. W6 · Ch.7 output: interaction and joint test
Men: return to education = **0.539** dollars per hour for each year (female = 0)
Women: 0.5395 − 0.0860 = **0.4535**
Gap: educ:female = −0.086 with p = **0.407**, so no evidence that the return differs
Joint test: F = **33.51** with p < 0.001 rejects H_0 that men and women have the same wage line
Why: female and educ:female are almost collinear (correlation 0.96), so their separate t-tests are weak while **jointly** they clearly matter
Reading: women earn less, the data just cannot say whether it is the intercept or the slope (at the mean educ 12.56 the gap is −1.1985 − 0.086 × 12.56 ≈ −2.28)
Key results: 0.539 · 0.453 · 0.407 · 33.5 · jointly

22. W6 · Ch.7 output: Chow test by hand
Formula: $F = [(SSR_r - SSR_{ur}) \div q] \div [SSR_{ur} \div (n - k - 1)]$
Numbers: $[(5548.2 - 4737.8) \div \mathbf{3}] \div (4737.8 \div \mathbf{520}) = 270.1 \div 9.111 = \mathbf{29.65}$
Restrictions: q = 3, the female intercept shift and its two slope interactions, and df = 526 − 6 = 520
Conclusion: p < 2.2e-16, so **reject** H_0 that men and women share one wage equation
Assumption: the Chow test needs the **same error variance** in both groups (Ch.7)
Key results: 29.6 · 3 · 520 · reject · same variance

23. W6 · Ch.7 output: linear probability model
educ: one more year of education raises the probability of being in the labor force by 0.038, that is **3.8 percentage points**, ceteris paribus
kidslt6: each child under 6 lowers it by **26.2** percentage points
Weakness 1: fitted values run from −0.345 to 1.127, and 33 of 753 (16 below 0, 17 above 1) fall **outside** [0, 1]
Weakness 2: Var(u | x) = p(x)[1 − p(x)] depends on x, so the LPM is necessarily **heteroskedastic** and needs robust standard errors
Also: a constant effect cannot hold everywhere, since going from 0 to 4 young children would lower the probability by 4 × 0.262 = 1.05
Key results: percentage points · 3.8 · 26.2 · below 0 · heteroskedastic